

Advanced Architecture for Demand Sensing and Probabilistic Forecasting: Integrating Exogenous Intelligence in Industrial Manufacturing

Executive Overview

The modernization of industrial supply chains necessitates a paradigm shift in demand forecasting, transitioning from purely autoregressive historical models to dynamic, context-aware demand sensing architectures. For an industrial manufacturing enterprise such as DEHN SE, headquartered in Nuremberg, Bavaria, historical sales data provides an incomplete picture of future demand¹. DEHN SE specializes in surge protection, lightning protection, and safety equipment across critical, fast-evolving infrastructure sectors, including photovoltaics (PV), e-mobility, power grids, and process industries¹. Demand for flagship component lines like DEHNguard, DEHNventil, and BLITZDUCTOR is inherently volatile and heavily influenced by macroeconomic indicators, regulatory shifts, public infrastructure tenders, and renewable energy expansion rates³.

The central architectural question focuses on the optimal methodology for integrating forward-looking, external information into the numerical forecasting pipeline. An initial hypothesis proposed an additive intelligence system, combining a baseline historical forecaster with a parallel external monitoring system to yield a final forecast adjustment. An exhaustive review of state-of-the-art time-series research, foundation models, and forecasting literature up to 2026 indicates that this additive architecture is structurally inadequate. It fails to capture the complex, non-linear, and time-lagged cross-correlations between exogenous market drivers and endogenous sales dynamics⁶.

This report recommends discarding the additive hypothesis in favor of a unified, attention-based deep learning architecture: TimeXer (Time Series Transformer with eXogenous Variables) coupled with Conformalized Quantile Regression (CQR)⁶. In this architecture, a Large Language Model (LLM) is strictly excluded from acting as the numerical forecaster due to proven limitations in native numerical sequence modeling and vulnerability to pseudo-alignment¹¹. Instead, the LLM is repositioned strictly as an asynchronous feature extractor in the data engineering pipeline, translating unstructured text from market news, construction tenders, and regulatory updates into structured time-series covariates¹³.

This document provides a comprehensive evaluation of contemporary forecasting paradigms, details the mathematical formulation of the recommended TimeXer-CQR architecture, outlines data requirements tailored to the German and European energy infrastructure market, and provides a rigorous validation and production roadmap. The analysis distinguishes between peer-reviewed evidence, preprints, experimental ideas, and the final architectural conclusion to ensure a robust, production-ready system design.

Industrial Context and the Demand Sensing Imperative

Industrial manufacturing supply chains operate under strict constraints regarding lead times, component procurement, and production capacity. Traditional forecasting systems rely on autoregressive integrated moving average (ARIMA) models, exponential smoothing, or basic machine learning approaches like LightGBM¹⁵. These models assume that the future is a linear continuation of the past. In stable markets, such techniques yield acceptable baseline forecasts. However, for a company embedded in the energy transition, the underlying data generating processes are fundamentally non-stationary.

DEHN SE's product ecosystem is tied to physical infrastructure deployment. When the German government alters subsidies for battery storage systems or mandates new lightning protection standards (e.g., IEC 62305 updates) for commercial buildings, the demand elasticity for specific SKUs shifts violently¹⁷. The concept of demand sensing emerged to address this latency between macroeconomic shifts and factory floor execution¹⁹. Demand sensing seeks to reduce demand latency by incorporating high-frequency downstream data and exogenous market signals directly into the forecasting algorithm¹⁹.

The challenge lies in translating theoretical demand sensing into a mathematical model capable of digesting hundreds of heterogeneous external signals without collapsing under computational complexity or succumbing to spurious correlations.

Critique of the Preliminary Additive Architecture Hypothesis

The proposed preliminary architecture suggests using a base model to forecast historical sales,

denoted as a function $f(X_{endo})$, and a parallel intelligence system to estimate the effect of external developments, denoted as $g(X_{exo})$, yielding a final forecast of $f(X_{endo}) \pm g(X_{exo})$.

While conceptually intuitive from a human reasoning perspective, this decoupled additive approach suffers from severe mathematical and operational limitations within the context of machine learning and supply chain dynamics:

Failure to Capture Multiplicative and Lagged Effects

External events rarely impact supply chains in a purely additive manner. For instance, a surge in national grid infrastructure investment does not simply add a fixed, absolute quantity of sales to the DEHNshield or DEHNcombo product lines. Instead, it alters the fundamental elasticity of demand, creating a multiplier effect that cascades over several months⁵. An additive formulation assumes strict independence between the baseline trajectory and the external shock. If the base demand is zero due to seasonal closures, adding an absolute value from the external monitor will produce nonsensical positive demand. True market dynamics are highly interactive, requiring multiplicative or complex non-linear combinations⁸.

Ignorance of Phase Shifts and Lead-Lag Dynamics

Supply chain dynamics are characterized by lead-lag relationships. An external signal, such as an increase in copper prices or a new municipal tender for EV charging stations, may pull demand forward, causing a temporary spike followed by a trough. An additive model cannot dynamically realign the temporal phase of the endogenous series based on an exogenous trigger. The interaction between variables must be learned temporally, allowing the model to understand that a regulatory announcement today impacts demand precisely three months from now, rather than instantaneously²¹.

Error Propagation and Baseline Divergence

If the base model forecasting from historical sales fails to anticipate a structural break or a regime change, its baseline prediction will be fundamentally incorrect. Adding an adjustment to a fundamentally divergent baseline compounds the error variance, leading to catastrophic inventory misallocations²⁴. A robust system must allow external variables to influence the core representation of the endogenous sequence *before* the final projection is made, ensuring that the model understands the structural break intrinsically⁶.

Modern forecasting literature overwhelmingly dictates that endogenous patterns (historical sales) and exogenous drivers (market data) must be modeled jointly within a unified representation space, allowing the neural architecture to learn the intricate cross-variate correlations and conditional dependencies directly⁶.

Independent Review of Forecasting Paradigms

To construct a reliable production system, it is necessary to evaluate the spectrum of available forecasting paradigms. The landscape of time-series forecasting has evolved rapidly between 2023 and 2026, transitioning from recurrent neural networks to transformer-based architectures and, more recently, to foundation models and LLM adaptations.

Large Language Models (LLMs) as Zero-Shot Forecasters

The period between 2023 and 2025 saw a massive influx of research attempting to adapt pre-trained LLMs, such as GPT-4 or LLaMA, for time-series forecasting. Frameworks like Time-LLM and Amazon's Chronos treated time series as a foreign language²⁶.

In the Chronos framework, continuous numerical values are tokenized into discrete text tokens through scaling and quantization²⁸. These tokens are fed into a transformer-based language model (e.g., T5), which predicts subsequent tokens using a standard cross-entropy objective²⁸. Chronos-2, released in late 2025, expanded this by introducing group attention mechanisms for covariate-informed forecasting, aiming to serve as a universal, zero-shot foundation model³⁰. Similarly, Time-LLM utilizes a reprogramming framework, embedding time-series patches into text prototypes and using prompt-as-prefix techniques to guide the frozen LLM's reasoning²⁶. Despite the initial enthusiasm and strong benchmark performances reported in these papers, rigorous ablation studies published in top-tier venues (such as NeurIPS 2024 and ICML 2025) have fundamentally challenged the efficacy of LLMs for pure numerical forecasting¹¹. Research by Tan et al. demonstrated that removing the LLM component entirely, or replacing it with basic attention layers, often resulted in equal or superior forecasting performance on the same

datasets¹¹. The findings indicate that:

- LLMs struggle to inherently grasp temporal frequency, seasonality, and magnitude variations when forced into tokenized spaces¹¹. The order-independent nature of standard text transformers leads to a loss of critical time-based information.
- The massive parameter counts, ranging from 120 million to over 700 million parameters for models like Chronos-2, introduce unacceptable inference latency and computational overhead for enterprise supply chain execution, where thousands of SKUs must be forecast frequently¹².
- LLM forecasters frequently exhibit "pseudo-alignment," where they memorize patterns or rely heavily on the tokenization/detokenization logic rather than learning the true underlying dynamical system of the time series¹¹.

Based on peer-reviewed evidence, utilizing a massive LLM as the core numerical forecaster is rejected for this production system. The computational cost vastly outweighs the marginal, if any, predictive benefit over dedicated time-series architectures.

Pure Multivariate Transformer Architectures

The success of transformers in natural language processing naturally led to their adoption in time series. Models like PatchTST demonstrated that segmenting time series into non-overlapping patches (subseries) and processing them with self-attention drastically improved performance over point-by-point models³⁶. Patching preserves local semantic meaning and reduces the sequence length, mitigating the quadratic memory complexity of standard attention mechanisms.

However, standard multivariate transformers like iTransformer or Crossformer struggle with the specific problem of exogenous variables³⁶. In standard multivariate forecasting, all variables are treated equally; the model attempts to predict the future values of every input variable simultaneously⁶. In the industrial context, DEHN SE does not need to predict the future price of copper or the future number of government tenders; it only needs to predict its own sales⁶. Treating exogenous variables as equal targets introduces massive computational overhead and forces the model to learn unnecessary interactions, diluting its focus on the primary endogenous target⁶.

Covariate-Specific Deep Learning Architectures

A specialized subset of models has been developed specifically to handle exogenous variables (covariates) efficiently.

Temporal Fusion Transformer (TFT): Introduced by Lim et al., TFT is highly regarded in industry for its interpretability and robust handling of static metadata, historical covariates, and known future inputs³⁹. The architecture utilizes Variable Selection Networks (VSNs) to weigh the importance of different inputs dynamically, and Gated Residual Networks (GRNs) to suppress irrelevant features, allowing the model to adapt its depth based on data complexity⁴⁰. TFT has been widely deployed for energy and retail demand forecasting¹⁵. However, TFT processes covariates sequentially point-by-point. When scaling to dozens or hundreds of high-frequency external indicators, the computational burden becomes prohibitive⁷.

Time-series Dense Encoder (TiDE): An MLP-based model that utilizes multi-layer perceptrons

in an encoder-decoder structure to handle long-term forecasting with exogenous inputs⁴⁴. While computationally efficient and fast, TiDE lacks the sophisticated attention mechanisms required to isolate complex, shifting cross-correlations between varying external events and sales patterns⁸.

TimeXer: Introduced at NeurIPS 2024, TimeXer represents a structural breakthrough for forecasting with exogenous variables¹⁰. Developed by researchers at Tsinghua University, TimeXer explicitly separates endogenous targets (company sales) from exogenous variables (external intelligence)⁶. It utilizes patch-wise self-attention to capture the temporal dynamics of the sales data and variate-wise cross-attention to inject external knowledge⁶. Crucially, TimeXer scales linearly with the number of exogenous variables, making it computationally vastly superior to TFT when integrating massive amounts of external economic and industry data¹⁰. TimeXer achieved state-of-the-art results across numerous benchmarks, proving its capability to reconcile endogenous and exogenous information seamlessly⁴⁵.

Probabilistic Forecasting and Uncertainty Quantification

A point forecast is inherently inadequate for supply chain operations. To optimize safety stock, calculate reorder points, and balance holding costs against stockout risks, procurement teams require predictive distributions⁴⁹.

Many modern architectures, including TFT, utilize Quantile Regression optimized via the Pinball Loss function⁹. Pinball loss penalizes overestimation and underestimation asymmetrically, training the neural network to output specific quantiles (e.g., the 10th, 50th, and 90th percentiles)⁵⁴. While this provides a predictive interval, traditional quantile regression lacks rigorous mathematical guarantees regarding out-of-sample coverage under distribution shift⁹. The leading methodology for guaranteed uncertainty quantification is Conformal Prediction, specifically Conformalized Quantile Regression (CQR)⁹. CQR operates as a wrapper around any base predictive model. It calculates conformity scores on a hold-out calibration set and uses these scores to adjust the heuristic intervals produced by the base model⁹. The resulting conformal intervals carry a theoretical, distribution-free guarantee: a 90% conformal interval will contain the true future value exactly 90% of the time, regardless of the underlying data distribution or model misspecification.

Architectural Conclusion

Based on the synthesis of peer-reviewed literature, the optimal production system is a composite architecture. The core representation learning and forecasting engine must be **TimeXer**, due to its superior handling of exogenous variables via dual-attention mechanisms. The output of TimeXer must be calibrated using **Conformalized Quantile Regression (CQR)** to provide guaranteed safety stock bounds for inventory management. Finally, the **Large Language Model (LLM)** is relegated strictly to an upstream data engineering role, performing asynchronous neurosymbolic feature extraction on unstructured text to generate numerical covariates for TimeXer.

This unified approach prioritizes demonstrated forecasting performance, mathematical rigor, robustness, and production feasibility over the experimental novelty of end-to-end LLM forecasters.

The Recommended Architecture: TimeXer with CQR and LLM Feature Extraction

The following sections detail the exact mechanisms of the recommended approach, explaining how external information mathematically influences the forecast, the specific role of the LLM, and the justification for this design over alternatives.

1. Mathematical Formulation of the TimeXer Core

The numerical forecast is generated by a process that distinctively embeds historical sales and external market data, fusing them via an endogenous global token⁵⁹.

Let the endogenous historical sales series for a specific DEHN product category be

$X_{endo} \in \mathbb{R}^{L \times 1}$, where L is the historical look-back window. Let the matrix of external market indicators be $X_{exo} \in \mathbb{R}^{L \times C}$, where C is the total number of external variables (e.g., copper prices, EV registrations, construction indices, and LLM-derived sentiment scores).

Step 1: Endogenous Patching and Embedding Processing long sequences point-by-point degrades transformer performance and loses local semantic context. Therefore, the

endogenous sequence X_{endo} is divided into non-overlapping patches (subseries)³⁶. The patched series is projected into a high-dimensional embedding space:

$$H_{endo} = \text{Embed}_{patch}(X_{endo}) \in \mathbb{R}^{P \times d}$$

where P is the number of patches and d is the embedding dimension. A crucial innovation in TimeXer is the introduction of a learnable global token $g \in \mathbb{R}^{1 \times d}$. This token is appended to H_{endo} , serving as the central information hub that will bridge the endogenous and exogenous data streams⁴⁸.

Step 2: Exogenous Variate Embedding Unlike the endogenous series, which is patched temporally, the exogenous variables are embedded along the variate dimension. The entire time-history of a single external variable is compressed into a single, global token representing that variable's macroscopic state⁶.

$$H_{exo} = \text{Embed}_{variate}(X_{exo}) \in \mathbb{R}^{C \times d}$$

This step allows the model to assess the overall trend of an external variable (e.g., the accelerating trajectory of solar installations) rather than matching it point-by-point to sales

data, drastically reducing computational complexity from $O((LC)^2)$ to $O(C)$ ¹⁰.

Step 3: Variate-wise Cross-Attention (Injecting External Intelligence) This mechanism dictates exactly how the external information mathematically influences the numerical forecast.

The global token g queries the exogenous embeddings H_{exo} through a cross-attention mechanism⁴⁷.

Let the query be derived from the global token: $Q = gW_Q$.

Let the keys and values be derived from the exogenous covariates: $K = H_{exo}W_K$, and $V = H_{exo}W_V$.

The updated global token, now enriched with external market intelligence, is computed as:

$$g_{new} = \text{Softmax}\left(\frac{QK^T}{\sqrt{d}}\right)V$$

In this equation, the attention matrix explicitly calculates the relevance of each external variable to the target sales data⁴⁷. If a specific regulatory change or a spike in copper prices exhibits a strong correlation with DEHN PV protection sales, the Softmax function automatically assigns a high attention weight to that variate, funneling that external signal into g_{new} ¹⁰. If an exogenous variable is pure noise, its attention weight trends toward zero, protecting the model from spurious correlations¹⁰.

Step 4: Patch-wise Self-Attention

The updated global token g_{new} , now carrying the distilled intelligence of all external factors, replaces the original g in the endogenous sequence. A standard Transformer self-attention layer is then applied across the temporal patches:

$$H_{final} = \text{SelfAttention}([H_{endo} \parallel g_{new}])$$

The final representation H_{final} seamlessly integrates the intrinsic temporal seasonality of the sales data with the macroeconomic shocks injected via g_{new} . A linear projection head maps H_{final} to the future forecast horizon $Y \in \mathbb{R}^{H \times 1}$ ⁶.

2. Conformalized Quantile Regression (CQR) for Inventory Optimization

To transition from point forecasts to operational supply chain metrics, the architecture utilizes Conformalized Quantile Regression (CQR)⁹.

Methodology	Output Type	Coverage Guarantee	Use Case
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Point Forecast (MSE)	Single Value	None	Financial reporting
Quantile Regression (Pinball Loss)	Interval	Heuristic / Uncalibrated	Basic uncertainty visualization ⁹
Conformalized Quantile Regression	Interval	Strict Mathematical Guarantee	Automated Safety Stock & Reorder Points [cite: 9, 62]

Mechanism of Action:

1. The TimeXer model is trained using the Pinball Loss function to output an initial lower and upper bound ($\hat{q}_{low}, \hat{q}_{high}$) corresponding to desired quantiles (e.g., 10th and 90th percentiles)⁹.
2. During model validation, an independent calibration dataset is passed through the model. Conformity scores s_i are calculated, measuring the signed distance of the true sales y_i from the predicted bounds:

$$s_i = \max\{\hat{q}_{low}(x_i) - y_i, y_i - \hat{q}_{high}(x_i)\}$$

3. The system identifies the $(1 - \alpha)(1 + 1/n)$ empirical quantile of these conformity scores, denoted as $\hat{Q}$, where n is the calibration set size and α is the target error rate⁹.
4. In production, the final prediction interval for future demand x_{new} is strictly calibrated as:

$$[\hat{q}_{low}(x_{new}) - \hat{Q}, \hat{q}_{high}(x_{new}) + \hat{Q}]$$

This guarantees that future observations will fall within the interval with a probability of at least $1 - \alpha$ ⁹. For DEHN SE, this is transformative. The lower bound informs minimal procurement requirements, while the upper bound dictates the maximum safety stock needed to prevent stockouts during supply chain shocks. During highly volatile market events, the CQR intervals automatically widen, providing a quantifiable risk metric for the procurement team⁹.

3. The Neurosymbolic Role of the Large Language Model (LLM)

In a departure from end-to-end multimodal forecasting concepts, this architecture explicitly restricts the LLM to the data engineering layer as an asynchronous feature extractor¹³.

Vast amounts of external intelligence are locked in unstructured text: government legislation,

industry reports, public tender documents, and news regarding grid expansions. The LLM (deployed via an API or a localized open-weight model like LLaMA-3) is tasked with reading these text streams, performing semantic reasoning, and converting them into structured, numerical covariates¹³.

Example Workflow for DEHN SE:

1. **Ingestion:** The LLM monitors scraping pipelines for public tenders for EV charging infrastructure in Bavaria or updates to the IEC 62305 lightning protection standards¹⁷.
2. **Extraction:** Through strict prompting and function-calling (JSON output), the LLM extracts key entities: Estimated_Budget, Projected_Completion_Date, Number_of_DC_Chargers, and Regulatory_Severity.
3. **Quantification:** The LLM generates a normalized, continuous EV_Infrastructure_Intensity_Score or Regulatory_Impact_Index for the given month²¹.
4. **Integration:** This numerical score is appended as a new column to the exogenous matrix X_{exo} . TimeXer then natively ingests this matrix, using cross-attention to determine if this new LLM-derived signal actually correlates with downstream sales²¹.

This approach capitalizes on the LLM's true strength (semantic reasoning and entity extraction) while relying on TimeXer and CQR for what they do best: robust, mathematically sound, and computationally efficient numerical forecasting¹³.

4. Why this Approach is Stronger than Alternatives

1. **Versus Additive/Fusion Models:** TimeXer learns the causal impact of external variables dynamically through cross-attention, allowing for multiplicative effects and phase shifts, rather than applying a naive post-hoc adjustment⁶.
2. **Versus LLM Forecasters (Chronos/Time-LLM):** Ablation studies prove LLMs offer minimal advantages for pure numerical prediction and suffer from excessive latency and pseudo-alignment¹¹. The proposed architecture avoids the overhead of billion-parameter models in the critical forecasting loop, ensuring production feasibility and rapid inference.
3. **Versus Traditional Multivariate Transformers (iTransformer/Crossformer):** Traditional models treat exogenous variables as targets to be predicted, wasting compute and diluting representations⁶. TimeXer's asymmetric architecture focuses all representational power on the endogenous sales variable, scaling linearly with the addition of external data¹⁰.
4. **Versus Standard Deep Learning (TFT):** While TFT is excellent, its sequential processing and variable selection networks become bottlenecks when handling hundreds of covariates⁷. TimeXer's variate-level embeddings bypass this limitation.
5. **Robustness Guarantee:** The inclusion of Conformalized Quantile Regression ensures that the model's uncertainty bounds are mathematically valid, directly solving the inventory optimization problem⁹.

5. Citations and Open-Source Implementations

The architecture is built upon rigorously peer-reviewed foundations with maintained

open-source repositories:

Component	Key Literature / Origin	GitHub Repository
TimeXer (Core Forecaster)	Wang et al., "TimeXer: Empowering Transformers for Time Series Forecasting with Exogenous Variables", <i>NeurIPS 2024</i> ¹⁰ .	https://github.com/thuml/TimeXer [cite: 65, 66]
Conformal Prediction (CQR)	Romano et al., "Conformalized Quantile Regression", <i>NeurIPS 2019</i> ⁵⁶ . Cordier et al., "Flexible uncertainty estimation... via the MAPIE library", <i>PMLR 2023</i> ⁶⁷ .	https://github.com/scikit-learn-contrib/MAPIE [cite: 68]
LLM Feature Extraction	Tan et al., "Are Language Models Actually Useful for Time Series Forecasting?", <i>NeurIPS 2024</i> (Ablation justification) ¹¹ .	Various (LangChain, ZenML, OpenAI API)
TFT / DeepAR (Baselines)	Lim et al., "Temporal Fusion Transformers for Interpretable Multi-horizon Time Series Forecasting", <i>Int. J. Forecasting 2021</i> ³⁹ .	https://github.com/sktime/pytorch-forecasting [cite: 41, 69]

Data Requirements for DEHN SE

To fully leverage the TimeXer architecture, a robust data pipeline must be established, tailored specifically to DEHN's product ecosystem across energy, PV, e-mobility, and building infrastructure¹.

Endogenous Data (Internal)

- **Historical Sales:** Monthly or weekly aggregated unit sales and revenue, segmented by product family (e.g., DEHNshield for residential PV, BLITZDUCTOR for data centers, DEHNventil for industrial facilities) and geographic region¹.
- **Supply Chain Metrics:** Order backlog, factory fulfillment lead times, and current inventory levels to contextualize fulfilled sales vs. raw demand.
- **Pricing & Product Lifecycle:** Historical pricing data, bulk discount events, and product

lifecycle markers (e.g., the introduction of Advanced Circuit Interruption (ACI) technology)⁴.

Exogenous Data (External Market Intelligence)

Because TimeXer scales linearly with exogenous variables, the system can ingest highly granular external data without degradation¹⁰.

Data Category	Specific Source / Indicator	Relevance to DEHN SE Product Lines
Renewable Energy Deployment	Marktstammdatenregister (MaStR) via Bundesnetzagentur. Monthly registration data for PV arrays, wind turbines, and battery storage ⁷² .	Directly drives demand for DEHNcombo (PV), DEHNguard (DC applications), and HVI lightning protection. The MaStR mandates registration within 1 month of commissioning, providing an exceptionally high-quality, near real-time exogenous variable representing downstream renewable capacity expansion ¹⁸ .
E-Mobility Infrastructure	Federal Motor Transport Authority (KBA) EV registrations; Federal Network Agency charging station maps.	Correlates with sales of DEHNgate, DEHNshield ZP, and AC/DC wall-box protection units ⁵ .
Macroeconomic & Industrial	Ifo Business Climate Index (Germany), Purchasing Managers' Index (PMI), Eurostat construction output.	Leading indicators for commercial building, hospital, and data center investments requiring comprehensive surge protection concepts ¹ .
Commodity Markets	Copper, aluminum, and stainless steel spot prices (LME).	Impacts raw material procurement costs (e.g., for earthing materials) and can forecast pre-emptive

		purchasing behavior by DEHN's distributors ¹⁸ .
Unstructured Intelligence	Regulatory updates (e.g., IEC 62305 standard revisions), state subsidies for battery storage, major infrastructure tenders.	Extracted via the LLM to form continuous sentiment/impact scores, acting as leading indicators for future compliance-driven sales ¹⁴ .

Validation Methodology Without Data Leakage

A critical failure point in applied machine learning for time series is data leakage—allowing the model to inadvertently access future information during training or calibration, resulting in overly optimistic evaluation metrics. To ensure the TimeXer-CQR system is robust, the validation strategy must enforce strict chronological boundaries.

Block Time-Series Cross-Validation with Embargo

Standard random k-fold cross-validation is mathematically invalid for time series due to temporal autocorrelation.

1. **Expanding Window Validation:** The data is split chronologically. The model is trained on a growing historical window and tested on the immediate subsequent period, simulating real-world passage of time.
2. **Data Partitioning:** The chronological split must accommodate four distinct phases: Training (TimeXer weights), Validation (hyperparameter tuning and early stopping), Calibration (calculating the CQR conformal scores), and Testing (final performance evaluation)⁵⁸.
3. **Embargoing (Purging):** To prevent look-ahead bias, an "embargo" period must be inserted between the training/validation sets and the test sets. If the forecast horizon is H months, at least H months of data must be purged between the end of the training set and the start of the evaluation set to sever temporal leakage.
4. **Real-Time Simulation:** The model is evaluated precisely as it would operate in production. If predicting Q3 sales, the model may only access exogenous and endogenous data timestamped up to the absolute end of Q2.

Technical Risks and Open Research Questions

While the recommended architecture represents the apex of current forecasting science, several technical risks require ongoing management:

Exogenous Data Degradation and API Drift

The model relies heavily on the quality and timeliness of external APIs (e.g., the MaStR registry or Eurostat). If an API changes format, or if data reporting is delayed, the exogenous matrix

X_{exo} will contain missing values or sudden structural drops.

- *Mitigation:* TimeXer handles missing exogenous variables gracefully through its cross-attention mechanism; if an external signal degrades to noise or drops out, the attention weights naturally attenuate, and the model falls back on the robust endogenous temporal patterns encoded in the patch-wise self-attention layers¹⁰.

Regime Shifts and Concept Drift

Industrial markets are subject to sudden structural breaks (regime changes), such as unforeseen supply chain crises, geopolitical events, or sudden subsidy cancellations. A model trained on a period of steady growth will fail during a sudden contraction.

- *Mitigation:* Integration of dynamic drift detection protocols. While TimeXer serves as the core, principles from recent research such as Dynamic-TMoE (ICML 2026) can be utilized to monitor distribution shifts⁷⁵. The system continuously calculates the Maximum Mean Discrepancy (MMD) between the historical training distribution and incoming real-time data streams⁷⁶. If the MMD exceeds a predefined threshold (indicating a regime change), the CQR intervals will naturally widen to reflect the new epistemic uncertainty, and an automated alert will flag the model for retraining⁹.

LLM Hallucination in Feature Extraction

The LLM might misinterpret a news article or tender document, generating a false positive signal for infrastructure investment.

- *Mitigation:* Rigorous prompt engineering, few-shot prompting, and grounding the LLM exclusively in highly curated, whitelisted data sources (e.g., official government portals, BNetzA press releases). The LLM output must be strictly typed (e.g., via Pydantic schemas) to ensure consistent mathematical ingestion, and human-in-the-loop auditing should be periodically applied to the extracted features.

Causal Confusion

With a vast number of external indicators, the neural network risks finding spurious correlations (e.g., linking a random weather variable in Asia to sales of DEHNguard in Bavaria).

- *Open Research Question:* Determining the optimal regularization penalty for the variate-wise cross-attention matrix to ensure that only causal, physically and economically grounded exogenous variables receive high attention weights. Exploring sparsity-inducing penalties on the attention matrix is an area for active research during model tuning.

Proof of Concept (PoC) Plan and Production Roadmap

Transitioning this architecture from theoretical research to a live production system for DEHN SE requires a phased Machine Learning Operations (MLOps) approach.

Phase 1: Data Infrastructure & LLM Extraction (Weeks 1-4)

- Establish secure data lakes on AWS or Azure to ingest DEHN's internal ERP data (SAP/Salesforce) and external APIs (MaStR, Eurostat, commodity prices).
- Deploy a lightweight LLM pipeline (using an orchestration framework like ZenML) to begin parsing regulatory updates and tenders, constructing the X_{exo} matrix⁷⁷.
- **Deliverable:** A unified, chronologically aligned dataset containing historical sales and

multi-variate exogenous features, mapped explicitly to product hierarchies.

Phase 2: TimeXer Proof of Concept (Weeks 5-8)

- Implement the TimeXer architecture using PyTorch. Reference the official implementation at <https://github.com/thuml/TimeXer>⁶.
- Select 3–5 high-volume, high-impact product lines (e.g., DEHNshield for residential PV, DEHNventil for industrial facilities) for the initial modeling scope⁵.
- Train the model to minimize Mean Absolute Error (MAE) and Mean Squared Error (MSE) over a 3-to-6-month forecast horizon.
- Establish Baseline Comparisons: Evaluate TimeXer against a purely historical ARIMA model, a standard XGBoost/LightGBM model, and the Temporal Fusion Transformer (TFT) using the pytorch-forecasting library (<https://github.com/sktime/pytorch-forecasting>) to empirically prove the superior value of the exogenous cross-attention mechanism¹⁵.
- **Deliverable:** Baseline validation metrics demonstrating superior point-forecast accuracy over legacy systems.

Phase 3: Uncertainty Quantification and Inventory Integration (Weeks 9-12)

- Integrate the MAPIE library (<https://github.com/scikit-learn-contrib/MAPIE>) to apply Conformalized Quantile Regression (CQR) over the TimeXer outputs⁵⁸.
- Translate the probabilistic bounds (e.g., the 90% prediction interval) directly into safety stock formulas. The upper bound of the CQR interval serves as the maximum inventory reorder point to prevent stockouts during supply chain shocks, while the median optimizes standard working capital⁴⁹.
- **Deliverable:** A dashboard displaying interactive forecasts with dynamic, mathematically guaranteed confidence intervals.

Phase 4: Productionization and MLOps (Weeks 13-16)

- Containerize the training, conformal calibration, and inference pipelines using Docker and Kubernetes.
- Implement MMD drift-detection monitoring to automatically trigger retraining when market regimes shift⁷⁶.
- Develop an inference API that feeds the dynamic safety stock parameters back into DEHN SE's enterprise resource planning (ERP) system for automated procurement and production scheduling.
- **Deliverable:** A fully automated, continuously learning demand sensing ecosystem ready for enterprise deployment.

Conclusion

The imperative to enhance historical sales forecasting with external market intelligence is highly sound; however, the initially proposed additive model architecture ($x \pm y$) is structurally inadequate for the non-linear, time-lagged complexities of global supply chain dynamics. By implementing the **TimeXer** architecture, DEHN SE can achieve deep, attention-driven fusion of endogenous sales history and exogenous market drivers, such as renewable energy

deployment via the MaStR registry and macroeconomic indices⁶. By restricting the **Large Language Model (LLM)** exclusively to the role of natural language feature extraction, the system sidesteps the latency, computational bloat, and mathematical instability inherent in end-to-end LLM-based numerical forecasters¹¹. Finally, by wrapping the deep learning predictions in **Conformalized Quantile Regression (CQR)**, the system transitions from merely outputting point-estimate guesses to providing strict mathematical guarantees on demand bounds⁹.

This composite architecture prioritizes robustness, interpretability, and direct operational utility, empowering DEHN SE to optimize working capital, minimize stockouts, and proactively adapt to the accelerating realities of the global energy transition.

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